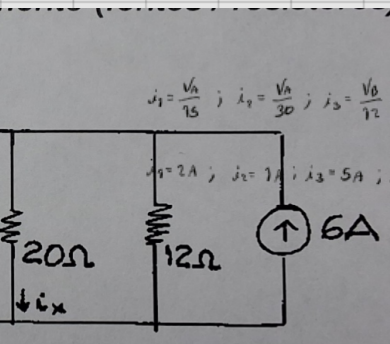




$$i_1 = \frac{V_A}{15}; i_2 = \frac{V_A}{30}; i_3 = \frac{V_B}{12}; i_X = \frac{V_B}{20}$$

$$i_1 = 2A; i_2 = 1A; i_3 = 5A; i_X = 3A$$

* Lei de Kirchhoff:

$$i_X = \frac{V_B}{20}; V_B - V_A = 10; i_X \rightarrow 2V_A - V_B = 0$$

$$V_B = 2V_A$$

$$V_A = 30V$$

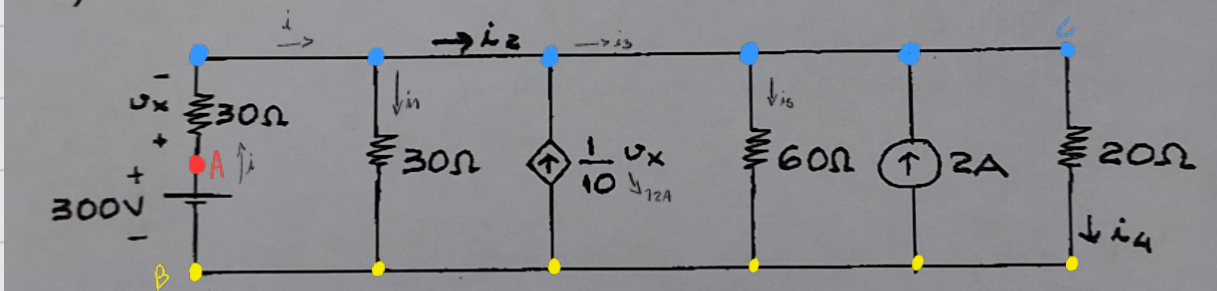
$$V_B = 60V$$

* Superfície A-B:

$$i_1 + i_2 + i_X + i_3 = 6 + 6$$

$$\frac{V_A}{15} + \frac{V_A}{30} + \frac{V_B}{20} + \frac{V_B}{12} = 12 \rightarrow 6V_A + 8V_B = 660$$

2)



→ Referência

$$V_A = +300V; i_1 = \frac{V_B}{30}; i_2 = \frac{V_B}{60}; i_3 = \frac{V_B}{20}; i_4 = \frac{300 - V_B}{30}$$

$$300 - V_B = V_X$$

$$V_X = 300 - 180$$

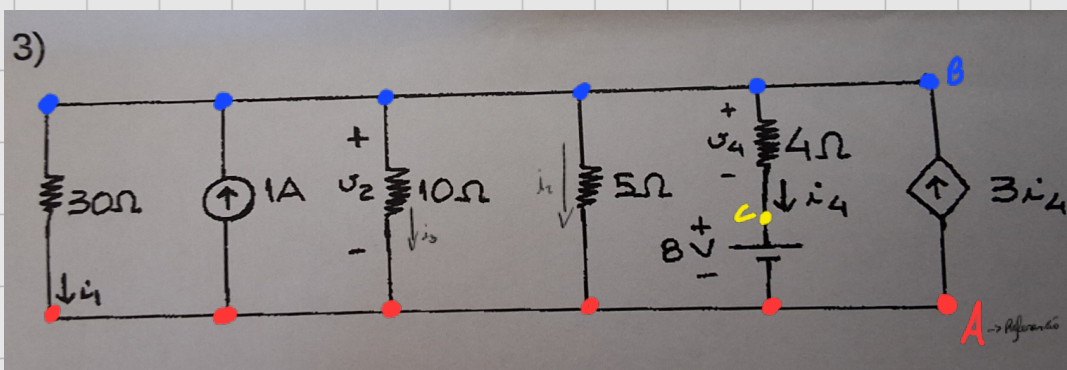
$$V_X = 120V$$

* NKC: $i_1 + \frac{3}{10}V_X + 2 = i_2 + i_3 + i_4$

$$10 - \frac{V_B}{30} + 30 - \frac{V_B}{70} + 2 = \frac{V_B}{30} + \frac{V_B}{60} + \frac{V_B}{20}$$

$$42 - \frac{V_B}{30} - \frac{V_B}{70} = \frac{V_B}{30} + \frac{V_B}{60} + \frac{V_B}{20} \times 70 \rightarrow 5040 - 4V_B - 22V_B = 4V_B + 16V_B + 7V_B \rightarrow 5040 = 28V_B \rightarrow V_B = 180V$$

3)



$$V_C = 8V; i_1 = \frac{V_B}{30}; i_2 = \frac{V_B}{5}; i_3 = \frac{V_B}{10}; i_4 = \frac{V_B - 8}{4}$$

* NKC: $i_2 + i_3 + i_4 + i_1 = 1 + 3i_4$

$$\frac{V_B}{30} + \frac{V_B}{10} + \frac{V_B}{5} + \frac{V_B}{4} - 2 = 1 + \frac{3}{4}V_B - 6$$

$$\frac{V_B}{30} + \frac{V_B}{10} + \frac{V_B}{5} + \frac{V_B}{4} - \frac{3}{4}V_B = -3 \times 60$$

$$2V_B + 6V_B + 72V_B + 75V_B - 45V_B = -180$$

$$-10V_B = -180 \rightarrow V_B = 18V$$

B

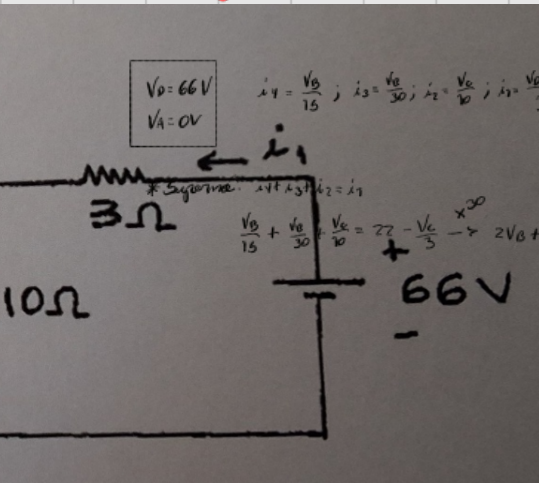


D $V_D = 66V$

A $\rightarrow$ Referência

i_3

i_2



$V_D = 66V$
 $V_A = 0V$

$$i_4 = \frac{V_D}{15}; i_3 = \frac{V_D}{30}; i_2 = \frac{V_C}{10}; i_1 = \frac{V_D - V_C}{3}$$

$$V_X = 10 \cdot i_2$$

$$i_1 = 2V_X$$

$$V_B - V_C = 2V_X$$

$$V_B - V_X = 2V_X$$

$$V_B = 3V_X = 3V_C$$

$$V_B = 90V$$

$$V_X = V_C$$

$$V_X = 30V$$

$$i_4 + i_3 + i_2 = i_1$$

$$\frac{V_D}{15} + \frac{V_D}{30} + \frac{V_C}{10} = 22 - \frac{V_C}{3} \quad \times 30$$

$$2V_B + V_D + 3V_C = 660 - 10V_C \rightarrow 6V_C + 3V_C + 3V_C = 660 - 2V_C \rightarrow 22V_C = 660 \rightarrow V_C = 30V$$